Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21311

F-411

B.C.A. (Third Semester) Examination, 2024-25

(New Course)

Paper - BCA-301-N

Computer Oriented Numerical Analysis

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. **Calculator** allowed.

1. (a) Define absolute, relative and percentage errors. An appropriate value of π is given by 3.1428571 and its true value is 3.1415926. Find the absolute and relative errors.

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(b) Solve the following:

(i) Prove that $(1+\Delta)(1-\nabla)=1$

(ii) Show that $E=(1-\nabla)^{-1}$

2. (a) Use Picard's method to obtain y for x=0.1, Given that : $\frac{dy}{dx} = x-y^2$; y=1 at x=0.

(b) Find the real root of $x^3-2x-1=0$ between 1 and 2 by bisection method. Compute five iterations.

3. (a) Find the cube root of 18 correct upto three decimal places by Regula falsi method.

(b) Evaluate $\int_3^7 x \log x dx$ taking 4 strips by simpson's ⅓ rd rule.

4. (a) Use the Runge Kutta method to approximate y for x=0.5 to x=1 using h=0.5, given that x=0 when y=1 and $\frac{dy}{dx} = \frac{1}{x+y}$

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(b) Evaluate $\int_0^6 \frac{1}{1+x^2} dx$ by simpson's 3/8th rule by dividing the interval into 6 parts.

5. (a) Find a real root of the equation $x^3-x-1=0$ using Newton Raphson method, correct to four decimal places.

(b) What are truncation errors in interpolation methods?

6. (a) Using Euler's method, find an approximate value of y corresponding to x=1, given that $\frac{dy}{dx} = x + y$ and y=1 when x=0

(b) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ correct to three decimal place using trapezoidal rule.

7. (a) Fit a straight line to the following data:

Year	1991	1992	1993	2994	1995	1996	1997	1998
Number	56	55	51	47	42	38	35	32

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(b) Find the two lines of regression for the data given below:

n=18, $\Sigma x=12$, $\Sigma y=18$, $\Sigma x^2=60$, $\Sigma y^2=96$, $\Sigma xy=48$.

8. (a) The population (in thousands) of a town in the years 1931, 1941, 1951, 1961 and 1971 are as follows:

Year	1931	1941	1951	1961	1971
Population	15	20	27	39	52

Find the population of the town in 1965 by applying Newton's backward formula.

(b) Explain Gauss Elimination methods with example.

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